

MAIE6000B GENERATIVE AI IN FINANCE

Homework #1

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Question 1: From Patterns to Generation

The transition from classical neural-network classification to generative models can be understood as a change in three things: what the model learns, how the training data is used, and what kind of output the model produces. In a classical classification task such as MNIST or Iris, the goal is to map an input to one label among a fixed set of categories. In a generative language model, by contrast, the goal is not to assign a label, but to model language well enough to produce a plausible continuation of text. Therefore, the move from classification to generation is not simply a matter of making a model larger. It is a shift from deciding among classes to modeling a distribution over possible next elements in a sequence.

In the classification setting, “similarity” means that a new input resembles previously seen examples in a learned feature space. For example, in the Iris dataset, a flower whose measurements are close to the cluster of *Setosa* examples should be classified as *Setosa*. In MNIST, a handwritten digit is mapped into an internal representation where images of the same number are placed relatively close together. The model learns decision boundaries that separate classes, so that similar inputs fall into the same region. In this setting, similarity mainly supports discrimination: the model asks which known class the new input is closest to.

The transformation from classification to generation happens in several conceptual steps. First, neural networks demonstrate that they can learn useful internal representations of patterns from data instead of relying on manually written rules. Instead of explicitly defining what makes a digit a “3” or what makes a flower belong to a species, the network learns such distinctions from many examples. Second, this representation-learning idea is extended from numerical and visual features to language. However, for language, similarity cannot be based only on surface appearance. Two words may look the same but mean different things in different contexts, while different words may be semantically related. This leads to the use of embeddings, in which words and later larger text units are represented as vectors in a high-dimensional meaning space. In this space, proximity reflects contextual and semantic similarity.

Third, once language is represented in this way, the objective changes. In classification, training is based on labeled pairs of the form

$$x \rightarrow y,$$

where x is the input and y is the class label. In a generative language model, the task becomes

$$(w_1, w_2, \dots, w_t) \rightarrow P(w_{t+1} \mid w_1, w_2, \dots, w_t),$$

which means that given the text so far, the model estimates a probability distribution

over the next token. This is fundamentally different from classification. The model is no longer learning only where to place a boundary between classes; it is learning the statistical structure of sequences, including syntax, semantics, and long-range contextual relationships.

A fourth step is that generation is recursive. A classification model usually produces one output and stops. By contrast, a generative model predicts one token, appends it to the sequence, and repeats the same process:

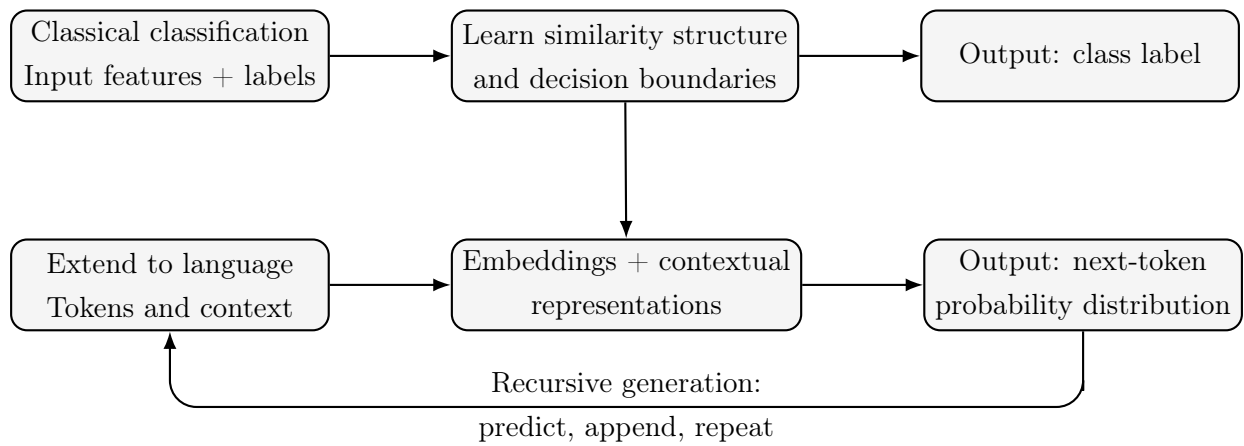
$$w_1, \dots, w_t \rightarrow w_{t+1}, \quad w_1, \dots, w_t, w_{t+1} \rightarrow w_{t+2}.$$

Repeating this process allows the system to produce full sentences, paragraphs, and documents. This is the operational step that turns a prediction model into a text generator.

Another major difference concerns data usage. Classical classification usually depends on labeled datasets. In MNIST, each image is paired with the correct digit. In Iris, each measurement vector is paired with a species label. Generative models, especially in pre-training, use data in a different way. Instead of relying primarily on manually labeled pairs, they learn from very large collections of raw text in a self-supervised manner. The text itself provides the training signal, because earlier tokens can be used as input and later tokens as targets. For example, in the sentence “The cat sat on the mat,” the prefix “The cat sat on the” can be used to predict “mat.” This makes it possible to learn from huge corpora such as books, articles, websites, filings, and reports without requiring manual label assignment for every example.

Overall, the path from classical neural-network classification to generative text modeling can be summarized as follows. First, neural networks learn internal representations that support similarity-based classification. Second, this representation-learning idea is extended to language through embeddings and contextual processing. Third, the objective is changed from predicting labels to predicting the next token in a sequence. Fourth, the prediction step is repeated recursively so that the model can generate new text. In short, classification asks, “Which class does this input belong to?”, whereas a generative language model asks, “Given this context, what should come next?” That change in objective and data usage is what takes us from pattern recognition to text generation.

Visual Schematic



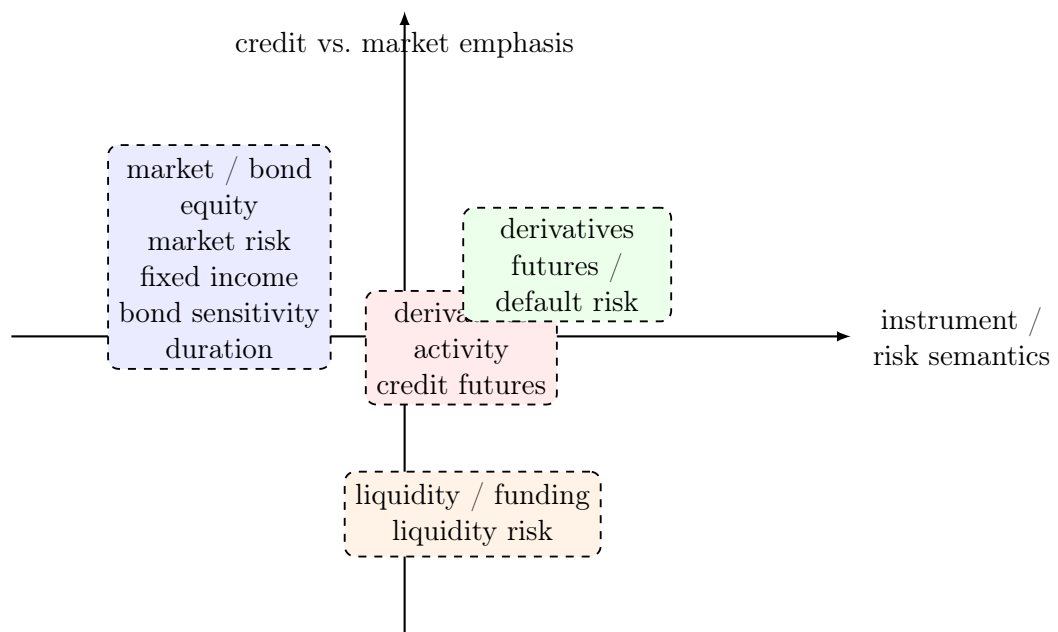
Short Comparison Table

Aspect	Classification Model	Generative Language Model
Main goal	Assign input to a known category	Generate plausible continuation of text
Similarity	Used to determine nearest class or region	Used in contextual meaning representations
Training data	Mostly labeled input-output pairs	Large-scale raw text, often self-supervised
Output	One label or class probability	Probability distribution over the next token
Inference style	One-step decision	Recursive sequence generation

Question 2: Building a “meaning space”

Using the idea of embeddings and similarity, we can imagine a two-dimensional “meaning space” for finance-related terms. The two axes below are only illustrative. They do not represent exact numerical dimensions learned by a real model, but they help visualize how semantically related terms can be placed close together while less related terms are placed farther apart. In the language of the lecture, the key idea is that embeddings map words into a space in which nearby items have similar meanings because they tend to appear in similar contexts.

Meaning-Space Sketch (2D)



Why certain terms are close together or far apart

The basic principle of this sketch is that terms placed near each other are expected to have similar meanings or to appear in similar financial contexts. This reflects the lecture idea of a *meaning space*, in which embeddings represent semantic proximity rather than simply visual or syntactic similarity.

First, *derivative*, *option*, and *futures* are placed close together because they belong to the same broad family of financial instruments. Options and futures are both types of derivatives, and they are often discussed together in contexts such as hedging, trading strategies, volatility, and risk exposure. Since they frequently co-occur in similar documents and explanations, their embeddings would be expected to lie near one another.

Second, *equity* and *market risk* are located relatively close together because equity prices are one of the most common sources of market risk. In practice, market-risk discussions often involve equity portfolios, price fluctuations, volatility, and trading losses. For that reason, these two terms share an important semantic neighborhood.

Third, *bond* and *duration* are placed close together because duration is a key concept in fixed-income analysis. It helps measure the sensitivity of bond prices to changes in interest rates. Even though duration is not itself a tradable asset, it is strongly linked to bond valuation and fixed-income risk management, so the two terms should appear near each other in a finance-related embedding space.

Fourth, *credit risk* is positioned near the derivatives and risk area, but not exactly on top of *market risk*. This is because credit risk concerns the possibility that a borrower or counterparty will fail to meet its obligations, whereas market risk is mainly about adverse price movements. The two are related, but they are not identical. Credit risk also connects to some derivatives contexts, especially when counterparty exposure is relevant, so it makes sense to place it between pure instrument terminology and broader risk terminology.

Finally, *liquidity risk* is placed farther away, in its own lower region. This is because liquidity risk usually refers to difficulties in funding, cash conversion, or the ability to trade without causing major price impact. It is related to finance generally, but it often appears in somewhat different contexts from equity trading, derivatives pricing, or fixed-income duration analysis. Therefore, it is reasonable to separate it from the main market-and-instrument clusters.

How such a space helps an LLM answer financial questions

A meaning space helps an LLM answer financial questions because it allows the model to represent words, phrases, and even larger pieces of text as vectors in a shared semantic space. When a user asks a question, the model does not only look for exact keyword matches. Instead, it can identify terms and concepts that are close in meaning.

For example, if a user asks how liquidity stress may affect derivative valuation, the model can connect *liquidity risk* with nearby concepts such as funding pressure, trading conditions, and market frictions, while also connecting *derivative* with pricing, hedging, and exposure. Because these concepts are represented in related neighborhoods, the model is better able to produce a coherent explanation that combines them.

This type of semantic structure also helps the model generalize beyond exact wording. A question may use “funding stress” instead of “liquidity risk,” or “shares” instead of “equity.” If the embeddings of these expressions are close to one another, the model can still interpret the question correctly and retrieve a relevant response pattern.

More broadly, such a space helps the model organize answers in a structured way. It can distinguish between nearby but different concepts, such as *credit risk* and *market risk*, while still recognizing that both belong to the broader domain of financial risk. In this sense, the meaning space acts as a semantic map that supports both understanding and generation. It helps the LLM produce answers that are more context-aware, more coherent, and more relevant to the financial topic being discussed.

Question 3: Pre-training vs Fine-tuning vs RLHF in Finance

When deploying a large language model (LLM) as a credit risk assistant in a bank, it is essential to distinguish between three training stages: pre-training, fine-tuning, and reinforcement learning from human feedback (RLHF). Each stage uses different data, teaches the model different capabilities, and introduces specific financial risks if done poorly.

1. Pre-training

Pre-training uses massive general-domain corpora such as books, websites, financial news, regulatory publications, and other publicly available text. The objective is typically next-token prediction:

$$(w_1, \dots, w_t) \rightarrow P(w_{t+1} \mid w_1, \dots, w_t).$$

At this stage, the model learns general language structure, reasoning patterns, financial terminology, and broad world knowledge. It does not yet specialize in a specific bank's credit policies.

If pre-training is done poorly (e.g., low-quality or biased data), the model may internalize misleading financial heuristics or biased risk narratives, which can later affect credit assessments in subtle ways.

2. Fine-tuning

Fine-tuning adapts the pre-trained model to a specific institutional context. For a credit risk assistant, the data may include internal credit policies, historical credit committee memos, anonymized loan case summaries, and regulatory guidelines (e.g., Basel frameworks).

At this stage, the model learns domain-specific language, institutional risk appetite, internal rating logic, and reporting formats. It becomes aligned with how the bank evaluates probability of default (PD), loss given default (LGD), and exposure at default (EAD).

If fine-tuning is done poorly, the model may misinterpret internal rating scales or reproduce outdated policy logic, leading to inconsistent or non-compliant recommendations.

3. Reinforcement Learning from Human Feedback (RLHF)

RLHF introduces human evaluators (e.g., risk officers, compliance experts) who rank or score model outputs. A reward model is trained to reflect preferred behavior, and the

LLM is optimized to produce outputs that align with these preferences.

At this stage, the model learns tone, prudence, escalation behavior, and when to express uncertainty. For example, it may learn to flag insufficient borrower information rather than fabricate assumptions.

If RLHF is poorly designed, evaluators may over-reward confident answers, unintentionally encouraging overstatement of credit conclusions. This can increase model risk and regulatory exposure.

Structured Overview

Stage	Data Used	Finance-Specific Risk if Poorly Done
Pre-training	Large-scale general text	Embedded bias or flawed financial reasoning
Fine-tuning	Internal policies, loan cases	Misalignment with bank risk framework
RLHF	Expert feedback	Overconfidence or unsafe recommendations

Question 4: Prompt Anatomy for an AML Monitoring Use Case

Selected Use Case: Anti-Money Laundering (AML) Transaction Review

Below is a full prompt structured into four components: instruction, context, constraints, and output format.

Full Prompt

Instruction: You are an AML compliance analyst. Review the following transaction summary and assess whether the activity is potentially suspicious.

Context: The customer is a small import-export company operating for 3 years. Monthly turnover averages USD 250,000. In the past 2 weeks, three incoming international transfers of USD 900,000 each were received from unrelated offshore entities.

Constraints: Base your assessment only on the provided information. Do not assume facts not stated. If information is insufficient, explicitly state what additional data would be required. Do not provide legal conclusions.

Output Format: 1. Risk indicators identified 2. Missing information 3. Preliminary risk assessment (Low / Medium / High) 4. Recommendation (e.g., escalate, request documents, monitor)

Annotation

The instruction defines the model's role to constrain reasoning. The context provides structured case facts. The constraints reduce hallucination and regulatory overreach. The output format enforces consistency and auditability, which are critical in compliance environments.

Question 5: Iterative Prompt Refinement

1. Experimental Design

Source material: HSBC Strategic Report 2025 (pp. 7–15).

Model used: GPT-5.4

Method: Identical source text used across all iterations. Prompts were progressively refined to improve analytical depth, neutrality, materiality prioritisation and risk awareness.

Objective: To evaluate how structured prompt engineering affects the quality of financial summarisation.

2. Iteration 0: Baseline Prompt

Prompt Used

Summarize the following HSBC Strategic Report 2025.

GPT-5.4 Output

HSBC Strategic Report 2025 — Summary

Overall message HSBC presents 2025 as a year of **strong financial performance, strategic simplification, and continued investment in core strengths**, particularly its international network, Hong Kong franchise, wealth business, and transaction banking capabilities. The narrative emphasises becoming a **simpler, more agile, higher-returning organisation**.

1. Financial performance

- **Reported profit before tax:** \$29.9bn
- **Profit before tax excluding notable items:** \$36.6bn
- **RoTE:** 13.3%
- **RoTE excluding notable items:** 17.2%
- **Revenue:** \$68.3bn
- **Dividend per share:** \$0.75
- **CET1 ratio:** 14.9%

Material shareholder returns included a fourth quarterly dividend of \$0.45 per share and \$3bn of share buy-backs.

2. Strategic direction HSBC reorganised into four core businesses:

1. Hong Kong
2. UK
3. Corporate and Institutional Banking (CIB)
4. International Wealth and Premier Banking (IWPB)

The structure is presented as enabling improved capital allocation and operational focus.

3. Simplification and restructuring A major theme is organisational simplification, including portfolio exits and cost discipline. The bank expects **\$1.5bn of annualised simplification savings** by end-June 2026.

4. Outlook and targets Medium-term targets include:

- Revenue growth of around 5% YoY (constant currency, excluding notable items)
- RoTE of 17% or better (2026–2028)
- Dividend payout ratio of 50%

Analysis:

The baseline prompt produced a broad narrative summary. While structurally clear, several limitations were observed:

- Marketing tone was retained.
- The output was descriptive rather than analytical.
- Limited prioritisation of material drivers.
- No discussion of downside risk or earnings sustainability.

3. Iteration 1: Financial Constraint Refinement

Prompt Used

Summarize the following HSBC Strategic Report 2025 in no more than 600 words. Focus strictly on:

- Key financial performance metrics
- Changes versus 2024
- Capital position
- Stated medium-term targets

Avoid marketing language. Use neutral analytical tone. Ensure all numerical figures are reported exactly as stated.

GPT-5.4 Output

HSBC reported **profit before tax of \$29,907m** in 2025, down from \$32,309m in 2024. **Reported revenue** increased to \$68,274m from \$65,854m. **RoTE** declined to 13.3% from 14.6%.

On a constant currency basis excluding notable items:

- **Profit before tax:** \$36,617m (vs \$34,181m)
- **Revenue:** \$71,020m (vs \$67,591m)
- **RoTE:** 17.2% (vs 15.6%)

Capital metrics:

- **CET1 ratio:** 14.9%
- **Total capital ratio:** 20.5%
- **Leverage ratio:** 5.3%

Medium-term targets (2026–2028):

- Revenue growth around 5% YoY
- RoTE excluding notable items of 17% or better
- Dividend payout ratio of 50%
- CET1 managed within 14%–14.5%

Analysis:

Improvements included removal of marketing language, systematic YoY comparison, and clearer separation of reported versus underlying performance. However, the output remained detailed and descriptive, with limited interpretation of performance drivers.

4. Iteration 2: Structural and Driver-Based Refinement

Prompt Used

Provide a 450-word analytical executive briefing on HSBC's 2025 performance.

Structure under:

1. Earnings performance and key drivers
2. Cost and efficiency trends
3. Credit risk and asset quality
4. Capital strength and shareholder returns
5. Medium-term outlook

GPT-5.4 Output

1. Earnings performance and key drivers

Reported profit before tax was **\$29.9bn**, down from \$32.3bn. Underlying profit (excluding notable items) rose to **\$36.6bn**. RoTE excluding notable items improved to **17.2%**.

2. Cost and efficiency trends

The cost efficiency ratio deteriorated to **53.4%**. Reported operating expenses increased 10% to \$36.4bn. Simplification savings of \$1.5bn are targeted by mid-2026.

3. Credit risk and asset quality

Credit impairment charges increased to **0.39%** of average loans (from 0.34%).

4. Capital strength and shareholder returns

CET1 ratio remained at **14.9%**. Dividend per share was \$0.75 with a 50% payout ratio.

5. Medium-term outlook

Targets include revenue growth of around 5% and RoTE of 17% or better.

Analysis:

Driver-based structuring improved coherence and materiality filtering. However, explicit downside analysis remained limited.

5. Iteration 3: Risk-Oriented Investor Briefing

Prompt Used

Provide a 450-word analytical investor briefing.

Identify:

- Downside risks to 17% RoTE
- Earnings sensitivities
- Distinction between reported and underlying results

GPT-5.4 Output

Reported RoTE was **13.3%**, while underlying RoTE reached **17.2%**. The divergence reflects notable items including impairment and restructuring costs.

Sustainability of the 17% target depends on three sensitivities:

1. **Margin pressure:** lower rate environment may compress spreads.
2. **Cost execution risk:** delivery of \$1.5bn savings is critical.
3. **Credit cost sensitivity:** impairment ratio increased to 0.39%.

CET1 of 14.9% supports capital returns, but sustaining distributions requires stable underlying profitability.

Analysis:

This iteration introduced explicit downside risks and earnings sensitivities, producing a research-note style output rather than a narrative summary.

6. Overall Reflection

Prompt refinement progressively increased analytical depth. Structural constraints improved clarity, but explicit risk requirements were necessary to elevate the output to investor-grade analysis.

The final iteration achieved the strongest balance between performance description and sustainability assessment.

Question 6: Human-in-the-Loop Verification Workflow

Use Case: Portfolio Recommendation Assistant

A verification workflow must address correctness, completeness, consistency, and compliance.

Stepwise Workflow

1. **Model Output Generation** The LLM generates a portfolio recommendation with asset allocation rationale.
2. **Automated Checks** - Correctness: Numerical consistency checks (allocation sums to 100%). - Completeness: Required fields (risk profile, time horizon) are present. - Consistency: Internal logic validation (e.g., high-risk client not assigned 90% bonds). - Compliance: Rule-based filters (e.g., restricted securities).
3. **Human Review** A licensed advisor reviews suitability, qualitative reasoning, and regulatory alignment.
4. **Final Approval and Logging** Decision and justification are logged for audit purposes.

Automation ensures scalability and error detection, while human intervention is required for fiduciary responsibility and regulatory accountability.

Question 7: RAG vs “Bare” LLM for Financial Questions

The difference between a bare LLM and a Retrieval-Augmented Generation (RAG) system is fundamentally about where knowledge resides and how it is accessed at inference time.

1. Knowledge architecture

A bare LLM relies entirely on parametric knowledge encoded in its trained weights. The information used to answer a question is implicitly stored during training and cannot be directly traced to a specific source. As a result, the model’s knowledge is effectively fixed after training and limited by its data cutoff.

In contrast, a RAG system separates knowledge storage from language generation. When a question is submitted, the system first retrieves relevant documents from an external corpus such as regulatory filings, research reports, or internal policy manuals. The LLM then generates a response conditioned explicitly on this retrieved context. Knowledge is therefore externalised rather than embedded.

2. Implications for financial use

This distinction has practical importance in finance. Regulatory standards, accounting rules, and internal risk policies change frequently. Updating a bare LLM typically requires retraining or fine-tuning, which is operationally costly and slow. A RAG system allows the institution to update its document repository without modifying the model itself.

RAG also improves traceability. Because responses are grounded in retrieved documents, institutions can log which sources informed a particular answer. This strengthens auditability and aligns more closely with regulatory expectations regarding explainability.

3. Redistribution of model risk

RAG mitigates certain weaknesses of bare LLMs. It reduces document-specific hallucination and helps address the problem of outdated embedded knowledge.

However, it introduces a different risk. If the retrieval process selects incomplete or irrelevant documents, the generated answer may still be misleading. In financial contexts, this shifts the primary risk from generative hallucination to retrieval quality and corpus governance.

RAG therefore does not remove model risk. It changes its location within the system architecture.

Question 8: Knowledge Graphs vs Chunked PDFs

Vector search over chunked PDFs works well for single-step information retrieval. However, financial supervision and due diligence often require reasoning across multiple entities and relationships. In such cases, knowledge graphs provide structural advantages.

1. Limits of chunk-based retrieval

Chunking divides documents into independent text segments that are indexed for semantic similarity search. While this allows flexible retrieval, it does not explicitly encode relationships between entities. If relevant information is distributed across separate documents, the system must rely on indirect semantic overlap rather than structured links.

This makes multi-step reasoning fragile. The model may retrieve only partial evidence or fail to connect related facts across documents.

2. Relational structure in knowledge graphs

A knowledge graph represents entities such as persons, banks, fintech firms, and regulatory authorities as nodes. Their relationships, such as employment, founding roles, or regulatory oversight, are represented as edges. This design encodes relational logic directly.

For example, if a person works at Bank X and also founded Fintech Y, and that fintech is regulated by Authority Z, the graph structure allows systematic traversal from one entity to another. The reasoning path is explicit rather than inferred from textual proximity.

3. Governance implications

In regulatory and investigative settings, tracing ownership, accountability, and potential conflicts of interest requires structured relational reasoning. Knowledge graphs improve completeness and transparency by making these links explicit.

However, constructing and maintaining a graph requires entity resolution, schema design, and ongoing data governance. The benefit in reasoning power must therefore be weighed against operational complexity.

In practice, combining vector retrieval for unstructured context with graph-based reasoning for entity relationships may offer the most robust architecture.

Question 9: Synthetic Data for a Financial LLM

Rare fraud detection presents a structural data problem. The most harmful events occur infrequently, which means that purely historical datasets often contain too few examples to train robust models.

1. Structural data imbalance

Fraud datasets typically exhibit extreme class imbalance. Fraudulent transactions represent a very small fraction of total activity, and privacy constraints further limit data sharing. Many edge-case fraud typologies appear only a handful of times. As a result, models trained solely on empirical data may perform well on common patterns but poorly on rare, high-impact scenarios.

2. Requirements for effective synthetic data

Synthetic data can help address this limitation by generating additional examples that preserve the statistical and behavioural structure of genuine fraud cases. However, synthetic samples must satisfy three conditions. They must be statistically realistic, diverse across fraud typologies, and designed to protect privacy so that no real individual can be identified.

If synthetic data introduces unrealistic correlations or simplified patterns, it may distort the model's decision boundary rather than improve generalisation.

3. Evaluation framework

The value of synthetic data should be tested empirically. A rigorous approach would train models on augmented datasets and evaluate them on held-out real-world data. Improvements in recall for rare fraud cases, without substantial loss in precision, would indicate practical benefit.

Synthetic data is therefore not simply a method for increasing dataset size. It is a targeted intervention designed to improve performance under structural scarcity.

Question 10: Alignment, Ethics, and Regulation in a GenAI Workflow

Responsible deployment of generative AI in finance requires a lifecycle perspective. Alignment and compliance are not achieved at a single stage but must be embedded throughout the system.

1. Lifecycle integration

The process begins with training data selection, followed by pre-training, fine-tuning, and reinforcement learning from human feedback. During deployment, prompt design shapes model behaviour at inference time. Human review mechanisms provide oversight before high-stakes outputs are released. Audit logging and continuous monitoring ensure ongoing compliance.

Alignment is therefore distributed across data governance, optimisation objectives, inference constraints, and post-deployment supervision.

2. Institutional safeguards

Prompt documentation and version control enhance traceability. Because prompts materially influence outputs, they should be treated as governed artefacts rather than informal instructions.

Mandatory human approval for high-stakes recommendations preserves fiduciary responsibility. The model provides analytical support, but legal accountability remains with licensed professionals.

Regular bias and fairness audits ensure that outputs do not systematically disadvantage particular demographic or risk groups. This requires structured monitoring rather than ad hoc review.

3. Redistribution of responsibility

Generative AI does not eliminate decision risk. It reshapes how risk is distributed between model, data, retrieval systems, prompts, and human supervisors. Effective governance combines technical safeguards with organisational accountability.

In regulated financial institutions, ethical deployment depends as much on institutional design as on model capability.